WEB PROGRAMMING LABORATORY
Mini Project Final Report

Academic Year: 2025–2026

Project Title:

ParkOS

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Course In-Charge:
Course incharge Name

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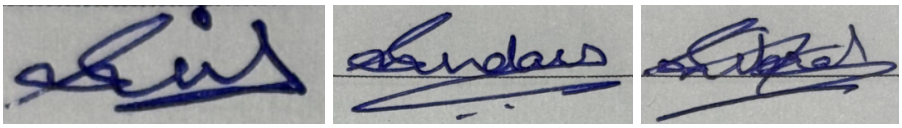
Declaration

This is to certify that the mini project titled '**ParkOS (A Smart Parking Management System)'** is a bonafide work carried out by the undersigned students under the guidance of the course instructors as part of the Web Programming Laboratory during the academic year 2025–2026.

Student Signature(s):



Date: 21/04/2026

Acknowledgement

We express our sincere gratitude to the Department of Computer Engineering, K. J. Somaiya School of Engineering, Somaiya Vidyavihar University for providing the opportunity to work on this mini project. We would like to thank our course instructor Shruti Javkar for their guidance and support.

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1. Introduction

Currently, there are challenges when it comes to managing parking spots in urban areas. With the increase in the population of vehicles in the city, traditional ways of managing parking spots have been inefficient because they lead to inefficiencies in time wastage and poor use of the available space. Therefore, it is necessary to develop a Vehicle Parking Management System to organize the parking activities.

Methodology:

The system we created uses HTML, CSS, JavaScript, PHP, and PostgreSQL technologies. The front-end part of the program will be user-friendly to ensure users have an easy time accessing the system. On the other hand, the back-end part will incorporate PHP to control application logic such as authentication, session control, and communication with the database. PostgreSQL is incorporated to handle all the information generated from the application.

Project Structure:

This project has several modules including user sign-up and login, administrative interface, creation of parking zones, management of parking sessions, slot allocation, and payment processing. The administrator is able to add parking zones and the number of slots. The users are required to enter vehicle details for registration purposes and park in a designated parking spot according to instructions provided by the system.

Overall, this project demonstrates how web development technologies can be combined to build a practical and useful application. It shows the importance of creating systems that are easy to use, efficient, and capable of solving real world problems like parking management.

2. Problem Statement

In today's fast-growing urban world, vehicle parking management becomes a challenging task since there are many vehicles. Conventional methods for parking are often inefficient and disorganized resulting in inefficiencies. The current work involves the development of Vehicle Parking Management System that ensures organized and efficient management of vehicle parking.

The web application is designed using languages such as HTML, CSS, Javascript, PHP, and PostgreSQL. The software supports several features including but not limited to user authentication, administration portal, parking zones and slots, parking sessions among others. Overall, the work showcases a good example of how web technologies can be utilized in building efficient solutions to real-world problems.

3. Objectives

- To develop a user friendly Vehicle Parking Management System.
- To provide an organized way to manage parking zones and slots.
- To enable users to register, login, and manage their vehicles.
- To allow administrators to monitor and control parking activities.
- To track parking sessions including entry and exit of vehicles.
- To improve parking space utilization and reduce manual effort.
- To design a responsive and interactive interface using HTML, CSS, and JavaScript.
- To implement server side functionality using PHP.
- To integrate database operations using PostgreSQL.

4. System Architecture / Web Graph

Home → Signup → Login → Dashboard → Manage Vehicles → Parking Zones → Book Slot / Start Parking → Parking Session → Payment → Logout

The user first lands on the Home page and can proceed to Signup to create an account, followed by Login for authentication. After logging in, the user is redirected to the Dashboard where they can manage their vehicles and view available parking zones. The user can then select a parking zone, choose an available slot, and start a parking session. The system tracks the parking session including entry and exit time, and once the session ends, the user can proceed with payment.

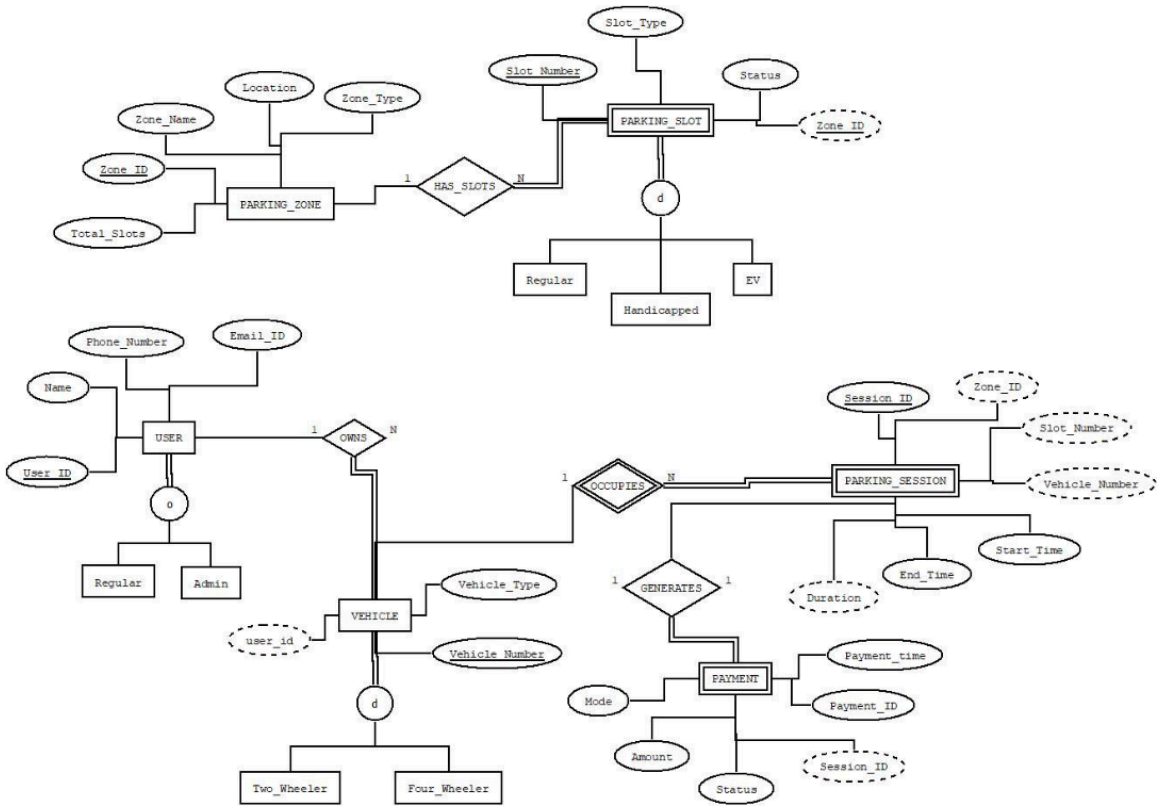
For administrators, after login, they are directed to the Admin Dashboard where they can create and manage parking zones, automatically generate slots, monitor slot availability, and view all parking sessions and payments. The system ensures smooth interaction between users and administrators while maintaining proper records of all activities.

5. Web Pages Developed

Page Name	Description
Home Page	Displays system introduction and basic navigation options.
Login Page	Allows users and administrators to log in using their credentials for authentication.
Signup Page	Allows new users to create an account and register their details.
User Dashboard	Provides an overview of user activities and access to different features.
Admin Dashboard	Allows administrators to manage zones, slots, users, and system data.
Manage Vehicles Page	Enables users to add and manage their vehicle details.
Parking Zones Page	Displays available parking zones and slot availability.

Manage Zones Page	Allows admin to create, update, and delete parking zones and generate slots.
Parking Session Page	Tracks active and past parking sessions including entry and exit details.
Payment Page	Handles payment details for completed parking sessions.

6. Database Design



Relational Model Schema:

1. Users:

Field	Type	Description
user_id	SERIAL (PK)	Unique ID for each user

name	VARCHAR	Name of the user
phone	VARCHAR	Contact number (unique)
email	VARCHAR	Email of the user (unique)
password	VARCHAR	Encrypted password
role	VARCHAR	Defines user role (admin or user)

2. Vehicle:

Field	Type	Description
vehicle_number	VARCHAR (PK)	Unique vehicle number
user_id	INT (FK)	References Users table

3. Zone:

Field	Type	Description
zone_id	SERIAL (PK)	Unique ID for each zone
name	VARCHAR	Name of the parking zone
location	VARCHAR	Location or description of zone
type	VARCHAR	Type of parking (covered, open, etc.)
total_slots	INT	Total number of slots in the zone

4. Slot:

Field	Type	Description
zone_id	INT (FK)	References Zone table
slot_number	VARCHAR (PK)	Slot identifier (e.g. ZA-01)
slot_type	VARCHAR	Type of slot (car, bike, truck)
status	VARCHAR	Availability status (available or occupied)

5. ParkingSession:

Field	Type	Description
session_id	SERIAL (PK)	Unique session ID
vehicle_number	VARCHAR (FK)	References Vehicle table
zone_id	INT (FK)	References Zone table
slot_number	VARCHAR (FK)	References Slot table
start_time	TIMESTAMP	Parking start time
end_time	TIMESTAMP	Parking end time

6. Payment:

Field	Type	Description
payment_id	SERIAL (PK)	Unique payment ID
session_id	INT (FK)	References ParkingSession table

payment_time	TIMESTAMP	Time of payment
amount	DECIMAL	Payment amount
mode	VARCHAR	Payment mode (cash, card, etc.)
status	VARCHAR	Payment status

7. GitHub Repository

https://github.com/SudarshanAchar/Parking_Lot_management-system.git

8. Comparative Analysis: Automated Coding Tools vs Manual Coding

Parameter	Manual Coding	Automated Coding Tools
Development Speed	Slower as code is written from scratch	Faster due to code generation and assistance
Code Accuracy	More number of errors in the code syntax and logically	Generally more accurate with almost no syntax errors
Learning Benefit	High, helps understand concepts deeply	Lower, may reduce conceptual understanding
Debugging	Requires manual effort and deeper analysis	Easier with suggestions and automated fixes
Maintainability	Better understanding leads to easier maintenance	Can be harder if code is not fully understood

9. Future Scope

● Mobile Friendly Interface – Enhance the design so the system works smoothly on phones and tablets for easier access.

● Online Slot Booking – Allow users to reserve parking slots in advance to reduce waiting time and congestion.

● Secure Authentication System – Improve user authentication with better security features like password encryption and session management.

● Payment Gateway Integration – Enable users to make online payments through integrated payment systems.

● Notification System – Provide alerts for parking status, session completion, and payments through SMS or email.

10. Conclusion

The Vehicle Parking Management System developed in this project provides an efficient and organized solution to manage parking operations. It simplifies tasks such as user registration, vehicle management, parking zone creation, slot allocation, and session tracking. The system reduces manual effort and improves the overall user experience by providing a structured and easy to use interface.

Through this project, the integration of front end and back end technologies has been demonstrated effectively. It highlights how web based systems can be used to solve real world problems in a practical way. Overall, the project serves as a reliable and scalable solution for modern parking management needs.

11. References

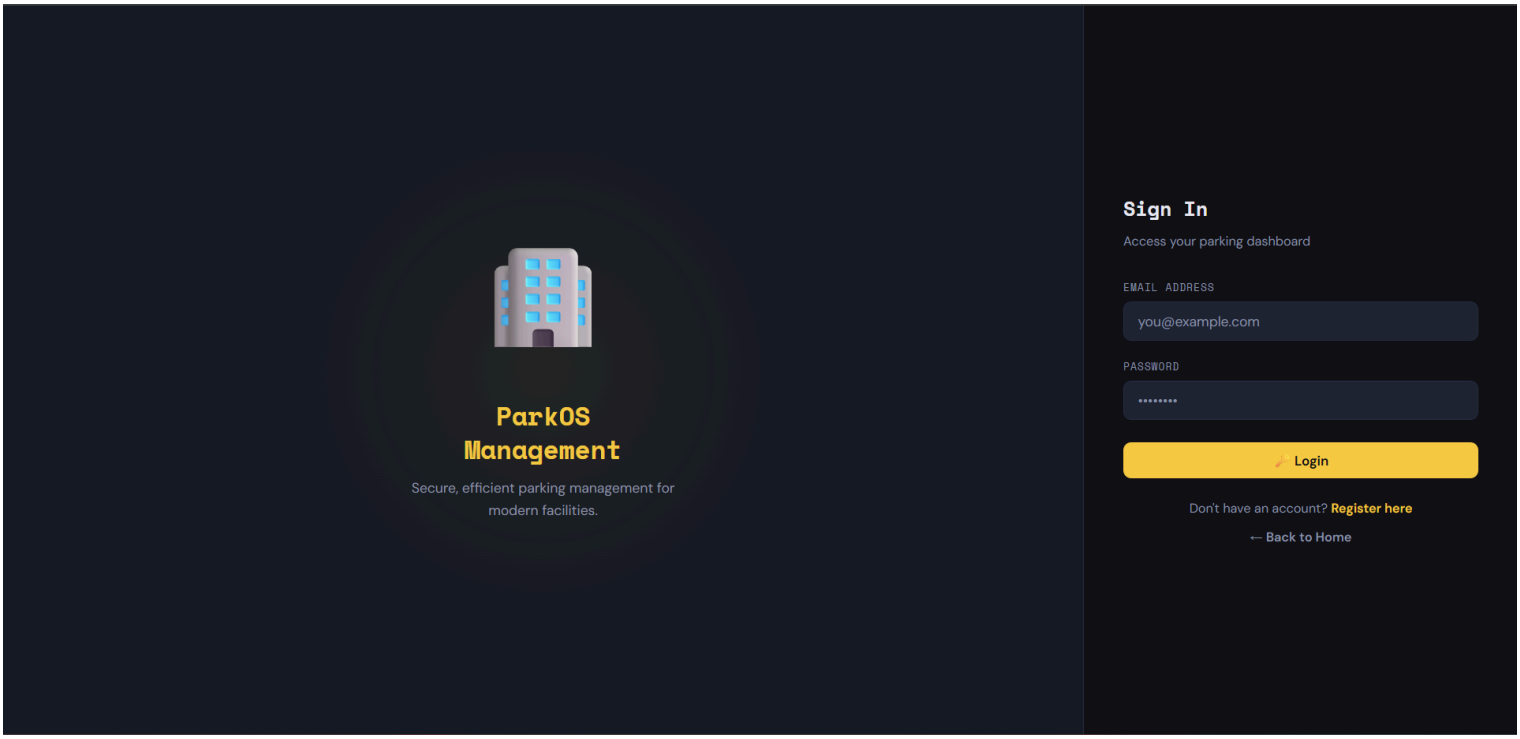
https://www.secureparking.co.in/

https://www.researchgate.net/publication/341870728_Smart_Parking_System_based_on_IOT?utm_source=chatgpt.com

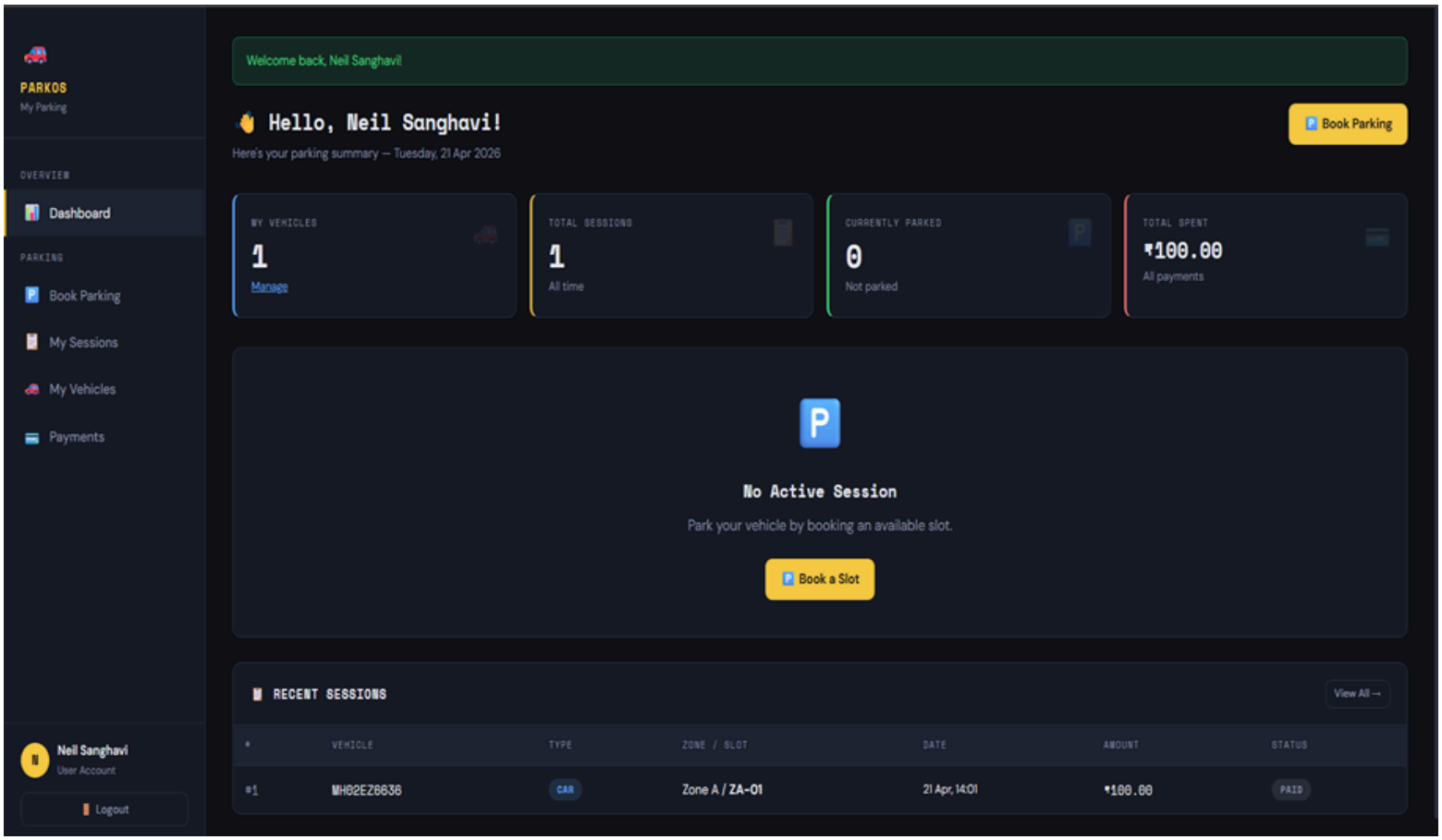
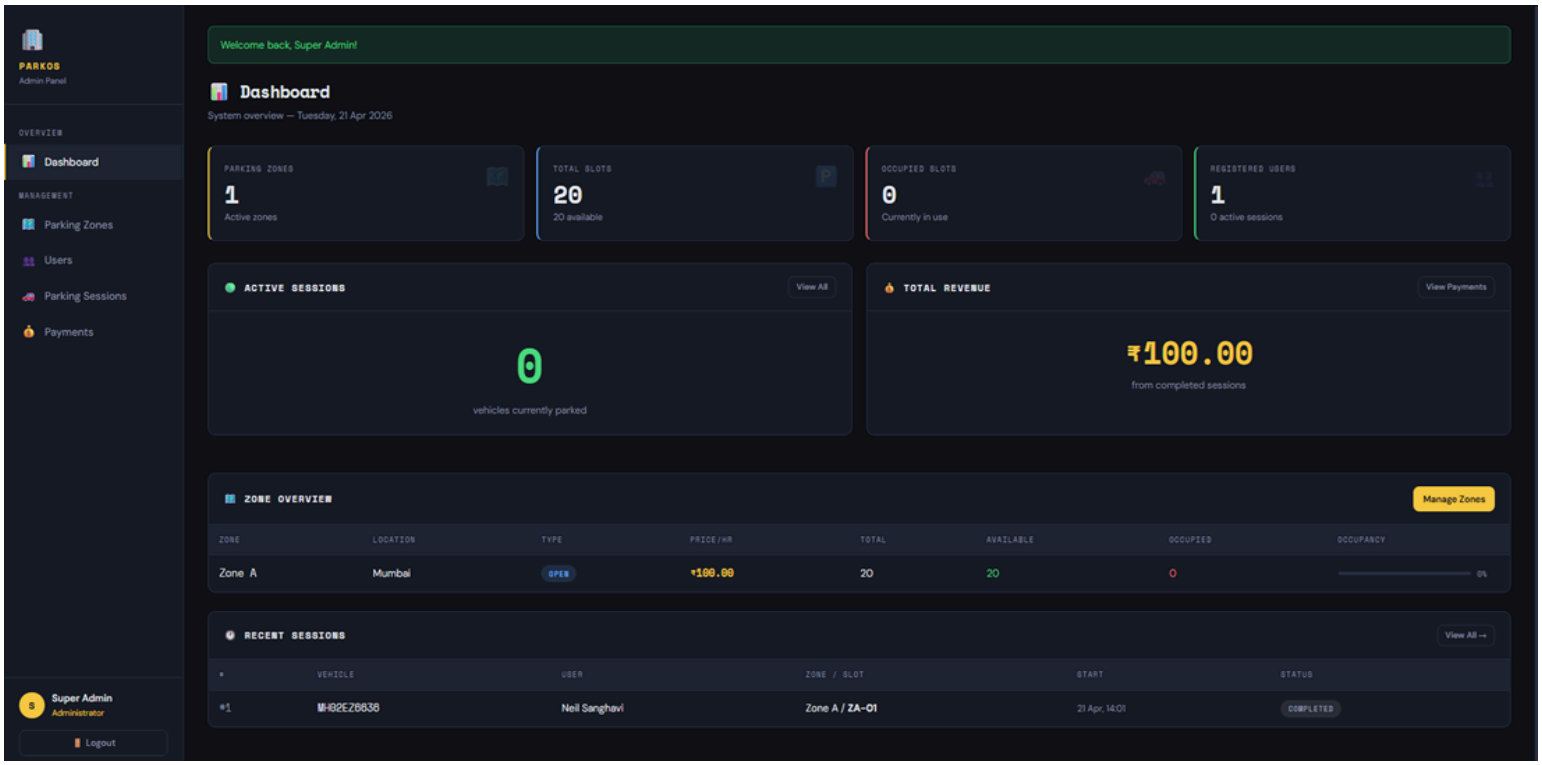
12. Appendix (Source Code / Screenshots)

Home Page:

Sign In Page:



User Page:



Admin Dashboard: